VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

Matunga, Mumbai – 400 019

Autonomous Institute Affiliated To University of Mumbai

EXAMINATION	Mid Semester Test	**DATE OF EXAM**	19/09/2024
SEMESTER & PROGRAM	Sem – III, SY.B.Tech.(CE/IT)	**TIME**	11:15 A.M. to 12:45 P.M.
TIME ALLOWED	90 minutes	**MARKS**	40
COURSE NAME – (CODE)	Mathematics for Computer Engineer (R5MA2007T)		

Instructions:	1.	All questions are compulsory.
	2.	All questions carry equal marks.
	3.	Figures to the right indicates full marks.
	4.	Answers to individual question must be grouped and written together.
	5.	Non-Programmable calculator is allowed.

Q.1	**b.**	Determine whether the given function: $f(z) = \begin{cases} \dfrac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ is analytic or not.	**(5)**	**CO5**
	c.	Show that the function $u = \log(x^2 + y^2) + x + y$ is harmonic. Find the corresponding conjugate harmonic function and construct the analytic function.	**(5)**	**CO5**
Q.2	**a.**	Determine the linear fractional transformation which maps the points $0, i, -1$ onto the points $1, 1+i, 0$ respectively. Hence find its fixed points.	**(5)**	**CO5**
	b.	Find image of the circle $\lvert z-2\rvert = 1$ under the bilinear transformation $w = \frac{z+1}{2z+3}$.	**(5)**	**CO5**

Q.3	**a.**	Solve the following system of linear equation using Crout's method: $x + 2y + z = 7$ $2x + 3y + 3z = 18$ $x + y + 2z = 10$	**(5)**	**CO1**
	b.	Find the dominant eigenvalue and corresponding dominant eigenvector of the matrix $A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$.	**(5)**	**CO2**
Q.4	**a.**	Evaluate: $\int_C (z^2 + z + 2)dz$, where C is the straight line joining the points $z = i$ to $z = 1$.	**(4)**	**CO6**
	b.	Evaluate the integral $\oint_C \frac{z^2+1}{z(2z-1)} dz$, where $C: \left\|z + \frac{1}{2}\right\| = 1$.	**(4)**	**CO6**
	c.	Show that $\oint_C \frac{z^2+1}{z^2(z-2)} dz = -\frac{\pi i}{2}$, where $C: \|z\| = 1$.	**(2)**	**CO6**

END OF PAPER

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EXAMINATION	Make-up Mid Semester Test	**DATE OF EXAM**	04/11/2024
SEMESTER & PROGRAM	Sem – III. SY. B. Tech.(CE/IT)	**TIME**	03:00 PM to 04:30 PM
TIME ALLOWED	90 minutes	**MARKS**	40
COURSE NAME – (CODE)	Mathematics for Computer Engineers (R5MA2007T)		

Instructions:	1.	All questions are compulsory.
	2.	All questions carry equal marks.
	3.	Figures to the right indicates full marks.
	4.	Answers to individual question must be grouped and written together.
	5.	Non-Programmable calculator is allowed.

Q.1	**a.**	Show that the function $f(z) = \sinh z$ is analytic and find its derivative.	**(4)**	**CO5**
	b.	Find the imaginary part of the analytic function whose real part is $e^{2x}(x\cos 2y - y\sin 2y)$. Express the analytic function in terms of z.	**(4)**	**CO5**
	c.	"A function $f(z) = r^3\cos k\theta + ir^k \sin k\theta$ is analytic." For which value of k, the given statement is true? Justify your answer.	**(2)**	**CO5**
Q.2	**a.**	Find the bilinear transformation under which $1, i, -1$ from the z −plane are mapped onto $0, 1, \infty$ of w −plane. Is there exist any fixed point under this transformation? If yes, find the fixed point.	**(5)**	**CO5**
	b.	Solve the following system of linear equation using Gauss Jacobi method: $2x - y = 7, -x + 2y - z = 1, -y + 2z = 1$	**(5)**	**CO5**

Q.3	a.	Solve the following system of linear equation using Doolittle method: $x + 2y + z = 7, 2x + 3y + 3z = 18x + y + 2z = 10$	(5)	CO1
	b.	Find the integral solutions of the equations: $252x + 198y = 18$	(3)	CO7
	c.	Show that square of any odd integer is of the form $8k + 1$.	(2)	CO7
Q.4	a.	Evaluate: $\int_C (z^2 + 3z^{-4})dz$, where C is the upper half of the unit circle from $(1, 0)$ to $(-1, 0)$.	(3)	CO6
	b.	Evaluate the integral $\oint_C \frac{\sin^6 z}{\left[z-\frac{\pi}{2}\right]^3} dz$, where $C: \|z\| = 2$.	(4)	CO6
	c.	Show that $\oint_C \frac{z^2+1}{z^2(z-2)} dz = -\frac{\pi i}{2}$, where $C: \|z\| = 1$.	(3)	CO6

END OF PAPER